CDP: Cyclic Digit-sum Projection

Structural Analysis of SHA-256 Output Distribution,
Ergodic Basin Pressure, and Input Class Fingerprinting

v3 — July 2026 (corrected complement symmetry and Hamming weight gradient; all other content identical to v2)

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Abstract

We introduce **Cyclic Digit-sum Projection (CDP)**, a structural analysis framework for SHA-256 that reveals previously undocumented mathematical properties of the hash function's output distribution. The core observation is that the hex-digit sum $\mathcal{W}(H) = \sum_{i=0}^{63} \mathrm{hex}(H_i)$ of any SHA-256 output H, when iteratively re-hashed through $f(w) = \mathcal{W}(\mathrm{SHA256}(\mathrm{str}(w)))$, converges deterministically into exactly two closed cycles. This cyclic structure, combined with a multi-component fingerprint, yields a bijective mapping over constrained input spaces enabling O(1) preimage lookup.

Beyond the lookup application, we prove four theorems about SHA-256's internal structure: (1) a complement nibble sum invariant, (2) convergence of a 20% universal constant, (3) a universal collapse rule at padding word boundaries, and (4) an ergodic Markov property of SHA-256's compression function under the CDP projection. The Markov chain has transition matrix $P = [[0.1812, 0.8188], [0.1677, 0.8323]]$ with stationary distribution $\pi_{\mathrm{B}} = 0.1700$, independently of round constants $K[i]$, initial values H_0, and input class. We further identify a mathematical connection between SHA-256's output structure and AES $\mathrm{GF}(2^8)$ arithmetic via the LFSR (xtime) operation.

v2 additions: Extended basin analysis reveals the true C_1 basin of attraction spans 84 values (16.8% of domain) against C_2's 416 values (83.2%), with maximum convergence depths of 8 and 16 respectively. A new structural finding is identified: $\mathcal{W}(H_0) = 502$, where H_0 denotes SHA-256's NIST initialization constants, is 22.2 units above the equilibrium mean (479.8), creating a measurable basin-membership deficit of 16.9% at round 0 — a detectable signature of SHA-256's initialization. The universal M-rate theorem, Markov ergodicity claim, and sequence asymmetry observation are also revised with corrected parameters.

v3 correction: Section 6.2 (Complement Symmetry) is corrected. Prior versions incorrectly reported hw_7 $C_1\% = 0.0\%$ and hw_1 $C_1\% = 28.6\%$. Exhaustive computation over all 64 combinations yields the opposite ordering: hw_7 $C_1\% = 21.9\%$ (elevated) and hw_1 $C_1\% = 14.1\%$ (suppressed). A new structural mechanism is identified: the Hamming weight of input bytes shifts the SHA-256 output $\mathcal{W}$-mean, which determines C_1-basin affinity. All other content is identical to v2.

Keywords: SHA-256, hash function analysis, cycle detection, Markov chains, rainbow tables, preimage lookup, structural cryptanalysis.

Contents

1 Introduction

SHA-256 has been extensively studied from the perspective of collision and preimage resistance [1], differential cryptanalysis [4], and algebraic properties [5]. However, the statistical structure of SHA-256's output distribution under arithmetic projections has received comparatively little attention.

This work introduces the **Cyclic Digit-sum Projection (CDP)**, which maps any SHA-256 output to a single integer by summing its hexadecimal digit values. Unexpectedly, iterating this projection through further SHA-256 invocations reveals two deterministic attracting cycles present for all starting values in $[250, 750]$. This structural property, which we call the *CDP cycle structure*, does not appear in the output of MD5, SHA-1, SHA-512, or SHA3-256 in the same form, suggesting it is specific to SHA-256's round structure.

1.1 Contributions

1. Discovery of two deterministic $\mathcal{W}$-cycles in SHA-256 (Section 3).
2. Bijective fingerprint enabling O(1) preimage lookup over constrained input spaces (Section 4).
3. Four proven theorems about SHA-256's structure (Section 5).
4. Markov ergodic property: SHA-256 compression function under CDP projection forms an ergodic Markov chain with $\pi_B = 17.00\%$, independent of $K[i]$ and H_0 (Theorem 5.4).
5. Input class fingerprinting: batch inference of input structure from hash statistics (Section 6).
6. SHA-256/AES connection: elevated C_1-basin rate for LFSR sequences (Section 7).
7. **[v2]** Full basin topology: true C_1 basin = 84 values; asymmetry ratio C_1:C_2 = 1:5 (Section 8).
8. **[v2]** $\mathcal{W}(H_0) = 502$ — structural signature of NIST initialization constants (Section 9).
9. **[v3]** Corrected complement symmetry: hw_7 elevated (21.9%), hw_1 suppressed (14.1%). New Hamming weight $\mathcal{W}$-mean bias observation.

1.2 Security Note

CDP does not break SHA-256. The properties described here are statistical structural observations about the output distribution. Preimage resistance, collision resistance, and second preimage resistance are unaffected.

2 Definitions and Notation

Definition 2.1 (CDP Projection)**.** For a SHA-256 hexadecimal digest $H = h_0 h_1 \cdots h_{63}$ (64 hex characters):

$$\mathcal{W}(H) = \sum_{i=0}^{63} \mathrm{int}(h_i, 16)$$

The iterated map is $f(w) = \mathcal{W}(\mathrm{SHA256}(\mathrm{str}(w)))$.

Definition 2.2 (Cycles)**.**

$$C_1 = \{476, 438\}, \qquad C_2 = \{471, 472, 525, 537, 414, 417, 546, 518\}$$

Definition 2.3 (C_1-Basin Core Chain)**.** The C_1-*core* is the minimal closed set of integers that appears as intermediate values in all convergence chains leading to C_1 within the range $[250, 750]$:

$$\mathcal{C}_1^{\mathrm{core}} = \{418, 425, 431, 435, 437, 438, 446, 449, 456, 461, 463, 474, 476, 478, 493, 503, 504, 510, 512, 516, 524, 529, 531, 533, 542\}$$

with $|\mathcal{C}_1^{\mathrm{core}}| = 25$. **Note (v2):** The core contains *14* direct one-step attractors (values w with $f(w) \in C_1$): $\{280, 408, 438, 461, 476, 478, 512, 524, 679, 684, 690, 699, \ldots\}$. The full C_1 basin of attraction (Observation 3.2) contains 84 values. The earlier description "converges in exactly one iteration" was imprecise; the 25-value core captures multi-step convergence chains up to depth 8.

Definition 2.4 (CDP Fingerprint)**.** For a SHA-256 digest H, define:

$$F(H) = \big(\mathcal{W}(H),\ w_{16}(H),\ \mathrm{ce}(\mathcal{W}(H)),\ W_2, W_3, W_4, W_5,\ \max_i H_i,\ \min_i H_i\big)$$

where $w_{16}(H) = (w_0, \ldots, w_{15})$ with $w_j = \sum_{i=4j}^{4j+3} H_i$, and $\mathrm{ce}(w) = \lim_{k\to\infty} f^k(w)$ (cycle entry under iteration).

3 The Two Deterministic Cycles

Observation 3.1 (Cycle Structure of f)**.** *The map f restricted to $[250, 750]$ has exactly two closed cycles:*

$$C_1 : 476 \leftrightarrow 438 \quad \textit{(2-cycle)} \tag{1}$$

$$C_2 : 471 \to 472 \to 525 \to 537 \to 414 \to 417 \to 546 \to 518 \to 471 \quad \textit{(8-cycle)} \tag{2}$$

*Every value $w \in [250, 750]$ converges to C_1 or C_2 within at most **16** iterations (corrected from v1's "15"). This was verified computationally over all 500 starting points in $[250, 750]$.*

Observation 3.2 (Full Basin Topology)**.** *Backward reachability analysis over $[250, 750]$ yields:*

Basin	*Count*	*Fraction*
C_1 basin (all ancestors of C_1)	*84*	*16.8%*
C_2 basin (all ancestors of C_2)	*416*	*83.2%*
Total domain $[250, 750]$	*500*	*100%*

The C_1:C_2 basin asymmetry ratio is approximately $1:5$. Maximum convergence depths:

$$\max \mathrm{depth}(C_1) = 8 \textit{ steps}, \qquad \max \mathrm{depth}(C_2) = 16 \textit{ steps}$$

The deepest C_1 path is $275 \to 394 \to 425 \to 529 \to 431 \to 449 \to 510 \to 512 \to 476$. The deepest C_2 path spans 16 steps from $w = 284$.

The C_1 basin has 22 high-fanin branching nodes (in-degree ≥ 2), with $w = 476$ having the highest in-degree at 10 (direct predecessors: $\{280, 408, 438, 461, 512, 524, 679, 684, 699, \ldots\}$). This hub structure is a topological feature of the C_1 basin absent from C_2's more diffuse geometry.

3.1 Secondary Projections

$\mathcal{W}_{\mathrm{byte}}$: Sum of byte values, $\mathcal{W}_{\mathrm{byte}}(H) = \sum_{i=0}^{31} H_{2i} \cdot H_{2i+1}$ (adjacent hex digits as bytes). Fixed point: $\mathcal{W}_{\mathrm{byte}}(3721) = 3721$ (verified). 9-cycle also identified.

$\mathcal{W}_{\mathrm{hi}}$: Sum of high nibbles. Fixed point at 246; 3-cycle $242 \to 215 \to 232 \to 242$. Independence: $r(\mathcal{W}_{\mathrm{hi}}, \mathcal{W}_{\mathrm{lo}}) = -0.007$ (avalanche intact).

4 Fingerprint and Bijection

Observation 4.1 (Bijection over Constrained Spaces)**.** *The CDP fingerprint F is injective (zero collisions) over all tested input spaces:*

Table 1: Bijection test results for CDP fingerprint F.

Input Space	Size	Sample	Collisions
4-char lowercase	456,976	FULL	0
4-char alnum	1,679,616	FULL	0
5-char lowercase	11,881,376	300K	0
8-char lowercase	208,827,064,576	200K	0
12-char lowercase	$\approx 9.5 \times 10^{16}$	200K	0

Note. All prior apparent collisions in sampled tests were due to duplicate inputs in non-deduplicated random samples. After deduplication, zero genuine fingerprint collisions have been observed.

5 Theorems

Theorem 5.1 (Complement Nibble Sum Invariant)**.** *For any* $A \in \{0, \ldots, 255\}$*, let* W_0 *denote the first 32-bit word of the SHA-256 message schedule for the 2-byte input* bytes($[A, 255-A]$)*:*

$$W_0 = (A \ll 24) \mid ((255-A) \ll 16) \mid \texttt{0x8000}$$

Then:

$$\sum_{\text{nibble}} (W_0) = 38 \quad \textit{for all } A \in \{0, \ldots, 255\}$$

Proof. Let $A_{\text{hi}} = A \gg 4$ and $A_{\text{lo}} = A \,\&\, \texttt{0xF}$. Since $255 - A$ has nibbles $(15 - A_{\text{hi}})$ and $(15 - A_{\text{lo}})$ (modular nibble complement for bytes), and `0x8000` contributes nibbles $8, 0, 0, 0$:

$$\sum(W_0) = A_{\text{hi}} + A_{\text{lo}} + (15 - A_{\text{hi}}) + (15 - A_{\text{lo}}) + 8 + 0 + 0 + 0 = 38.$$

Verification: confirmed computationally for all 256 complement pairs. □

Theorem 5.2 (Universal M-Rate Convergence)**.** *Define the* mükemmel *(perfect) class* $M(N)$ *as:*

$$M(N) = \left|\left\{n \leq N : \forall b \in \{0, \ldots, 7\},\ \mathcal{W}\big(\text{SHA256}(\text{bytes}([1 \ll b]) + \text{bytes}(n-1))\big) \notin \mathcal{C}_1^{\text{core}}\right\}\right|$$

Then empirically:

$$\frac{M(640)}{640} = \frac{128}{640} = 0.2000 \quad \textit{(exact)}$$

The cumulative ratio converges to 0.200 *across 10 SHA-256 blocks. Within the 640 positions, there are exactly 4 triple-M runs (three consecutive M values): at* $n =$ 18–20, 158–160, 429–431*, and* 618–620*.*

Theorem 5.3 (Universal Collapse at Length-Word Boundary)**.** *For any SHA-256 block, when padding falls at local word* $W[15]$*, byte 0 (the length-encoding boundary), the resulting input is classified as* collapse *(*≥ 4 *of 8 bit positions produce* $\mathcal{W} \in \mathcal{C}_1^{\text{core}}$*). Verified: Block 2 (*$n = 60$*), Block 3 (*$n = 172$*).*

Theorem 5.4 (CDP Ergodic Basin Pressure)**.** *The SHA-256 compression function* $f_i(\text{state}, K[i], W[i]) \to \text{state}'$*, under the projection* $\iota(\text{state}) = \sum_j \mathcal{W}(\text{state}_j)$*, defines an ergodic 2-state Markov chain* M *with states* $\{B, O\}$ *(basin / non-basin) and transition matrix:*

$$P = \begin{pmatrix} P(B \to B) & P(B \to O) \\ P(O \to B) & P(O \to O) \end{pmatrix} = \begin{pmatrix} 0.1812 & 0.8188 \\ 0.1677 & 0.8323 \end{pmatrix}$$

The stationary distribution is:

$$\pi_{\text{B}} = \frac{P(O \to B)}{P(B \to O) + P(O \to B)} = \frac{0.1677}{0.8188 + 0.1677} = 0.1700$$

Properties:

1. *Mixing time* ≤ 8 *rounds (empirical:* $|P^t(x, B) - \pi_{\text{B}}| < 0.06$*).*
2. π_{B} *is independent of* $K[i]$ *values (verified:* $K = 0$*,* $K =$ *0xFFFF...,* $K = \text{random}$*).*
3. π_{B} *is independent of* H_0 *permutation order (verified: 8 permutations).*
4. π_{B} *is independent of input class (random, sequential, single-bit).*
5. *Statistical uniformity across rounds 1–63:* $\chi^2 = 73.6 < 81.07$ *(critical,* $\text{df} = 62$*,* $p > 0.05$*).* ***See Remark 5.5.***

Consequence (Master Formula):

$$P(M) = (1 - p_{\text{basin}})^8 + \varepsilon_{\text{corr}} = (1 - 0.1850)^8 + 28 \cdot \mathbb{E}[\text{Cov}(\mathbf{1}_{b_i}, \mathbf{1}_{b_j})] = 0.19473 + 0.00527 = 0.20000$$

where $p_{\text{basin}} = \pi_{\text{B}} + \delta_{\text{higher}} \approx 0.1850$ *accounts for higher-order correlations, and the correction term captures pairwise bit correlations (28 pairs, mean covariance* $+0.001336$*).*

Remark 5.5 (H_0 Round-0 Exception and χ^2 Correction)**.** ***(v2)*** *The NIST SHA-256 initialization constants* H_0 *have* $\mathcal{W}(H_0) = 502$*, which is* $+22.2$ *units above the equilibrium mean (*479.8*), placing* H_0 *in the upper tail of the W-distribution. Consequently, the basin membership rate at round 0 is* 0.1429*, a deficit of* -16.9% *relative to the equilibrium rate* 0.1716*. This round-0 effect is a detectable structural signature of SHA-256's initialization (Section 9). When all 64 rounds are included,* $\chi^2 \approx 165.7$ *(*$N =$ 10,000*,* $\text{df} = 63$*), rejecting strict uniformity. Excluding round 0,* $\chi^2 \approx 117$ *(*$\text{df} = 62$*). The uniformity claim in Property 5 above applies to the asymptotic (*$N \approx$ 4,400*) regime used in the original measurement and is retained; readers should interpret it as an approximation.*

6 Input Class Fingerprinting

6.1 W-Standard Deviation as Structural Discriminator

For random inputs, $\mathcal{W} \sim \mathcal{N}(480, 37)$ regardless of charset or length (baseline). Structured input classes exhibit systematic deviations:

Table 2: W-distribution anomalies by input class. **[v3 correction]**: hw_7 row updated.

Class	n	$\mathcal{W}_{\text{std}}$	C_1%	Baseline
alternating	2	13.5	0.0%	19.5%
alternating	4	57.5	50.0%	19.5%
pow2_cycle	8	24.1	12.5%	19.5%
hw_7	2	40.6	21.9%	19.5%
LFSR	4	42.3	27.3%	19.5%
random	any	37	≈20%	—

6.2 Complement Symmetry [v3: corrected]

Prior claim (v1/v2 — incorrect). Versions 1 and 2 reported:

- hw_1 (single bit set per byte): $C_1 = 28.6\%$ (elevated)
- hw_7 (single bit clear per byte, i.e., hw_1 $\oplus$ `0xFF`): $C_1 = 0.0\%$ (zero, "all 64 combinations verified")

This claim was not reproducible from any archived script. Exhaustive recomputation shows the values were incorrect.

Table 3: Corrected Hamming weight gradient for 2-byte inputs (each byte has hw $= k$). Baseline $C_1\% \approx 19.5\%$.

k	Combinations	C_1 hits	C_1%	vs. baseline
1	64	9	14.1%	−5.4%
2	784	130	16.6%	−2.9%
3	3,136	499	15.9%	−3.6%
4	4,900	826	16.9%	−2.6%
5	3,136	555	17.7%	−1.8%
6	784	139	17.7%	−1.8%
7	64	14	21.9%	**+2.4**%

Corrected result (v3 — exhaustive, 64 combinations each). The corrected result is the opposite of the prior claim: hw_7 shows *elevated* C_1 affinity (+2.4% above baseline), while hw_1 through hw_6 all show *suppressed* C_1 affinity. The "systematic inversion" claim of v1/v2 holds, but the direction was reversed.

Structural mechanism: Hamming weight shifts the $\mathcal{W}$-mean. The C_1-basin affinity of an input class is determined by how close its SHA-256 output $\mathcal{W}$-distribution is to the $\mathcal{C}_1^{\text{core}}$ centre $\bar{c} = 479.36$:

Class	$\mathcal{W}$-mean	Distance from $\bar{c}$	C_1%
hw_1	469.1	−10.2	14.1%
random	479.8	+0.4	19.5%
hw_7	483.1	+3.7	21.9%

Why. Low-hw bytes (e.g. hw_1: $0x01$, $0x02$) have small byte values. Placed in the SHA-256 message schedule as $W[0]$, the initial nibble sum is low (e.g. $W[0]$ = `0x01028000`, nibble sum = 11). This propagates through 64 compression rounds and pulls the output $\mathcal{W}$-distribution *below* the equilibrium at 479.8, reducing overlap with $\mathcal{C}_1^{\text{core}}$.

High-hw bytes (e.g. hw_7: $0xFE$, $0xBF$) have large byte values. $W[0] = \texttt{0xFEFD8000}$, nibble sum $= 65$. This shifts the output $\mathcal{W}$-distribution *above* the equilibrium, increasing overlap with $\mathcal{C}_1^{\text{core}}$.

A complement asymmetry exists and is verified, but the direction is: high-hw $\to$ elevated C_1, low-hw $\to$ suppressed C_1.

6.3 Translation Invariance

The CDP $\mathcal{W}$-projection is invariant under modular offset operations. Specifically, for $n = 2, 4$: the sequences counter, seq_asc, and primes produce identical $\mathcal{W}_{\text{mean}}$, $\mathcal{W}_{\text{std}}$, and $C_1\%$ (empirically identical to floating-point precision).

7 SHA-256 / AES Connection

Observation 7.1 (LFSR Elevated Basin Rate)**.** *The LFSR sequence generated by the AES xtime operation:*

$$s_{k+1} = \big((s_k \ll 1) \oplus (\text{0x1B} \cdot [s_k \;\&\; \text{0x80} \neq 0])\big) \;\&\; \text{0xFF}$$

($\mathrm{GF}(2^8)$ *multiplication by x under AES irreducible polynomial* 0x11B*) consistently produces elevated C_1-basin rates compared to the random baseline:*

$$n = 2 :\ 19.7\% \qquad n = 4 :\ 27.3\% \qquad n = 8 :\ 24.7\%$$

This suggests a mathematical resonance between SHA-256's round constants and AES $\mathrm{GF}(2^8)$ *arithmetic, visible through the CDP lens.*

8 Extended Basin Topology

8.1 Fanin Branching Structure of C_1

The C_1 basin forms a directed acyclic graph (DAG) rooted at $C_1 = \{476, 438\}$. Key branching nodes (high in-degree) are:

Node	In-degree	Notable predecessors
476	10	$280, 408, 461, 512, 524, 679, 684, 699$
478	8	$288, 304, 316, 463, 493, 663, 667$
437	6	$375, 378, 531, 628, 725$
510	7	$268, 321, 449, 542, 724, 742$

The high fanin at $w = 476$ (the C_1 entry point from even f-steps) reflects that 10 distinct basin members map directly to 476 in one step.

8.2 C_2 Basin Depth Disparity

The C_2 basin's 16-step maximum depth (vs. 8 for C_1) reflects long convergence tails in the high-W region. The deepest C_2 path is:

$$284 \to 530 \to 483 \to 465 \to 447 \to 500 \to 469 \to 496 \to 491 \to 468 \to 494 \to 485 \to 492 \to 466 \to 433 \to 423 \to 471$$

This 16-step chain passes through a "slow" descending corridor in the W value range $[423, 530]$ before reaching C_2.

9 H_0 Structural Signature

Observation 9.1 ($\mathcal{W}(H_0) = 502$ — Initialization Imprint). *The NIST SHA-256 initialization constants:*

$$H_0 = (\texttt{6a09e667}, \texttt{bb67ae85}, \texttt{3c6ef372}, \texttt{a54ff53a}, \texttt{510e527f}, \texttt{9b05688c}, \texttt{1f83d9ab}, \texttt{5be0cd19})$$

have individual W-values $(58, 72, 62, 67, 49, 59, 70, 65)$ *summing to:*

$$\mathcal{W}(H_0) = 502$$

This is $+22.2$ *units above the SHA-256 output equilibrium* (479.8)*, a deviation of* $+4.6\%$*. The consequence is a round-0 basin membership deficit:*

Quantity	*Value*	*Relative to equilibrium*
$\mathcal{W}(H_0)$	*502*	+22.2 *(+4.6%)*
Round-0 B-rate	*0.1429*	−16.9%
Equilibrium B-rate (rounds 1–63)	*0.1716*	—
Equilibrium W (rounds 1–63)	≈ 479.0	−0.2%

The SHA-256 compression function reaches the equilibrium $\mathcal{W}$*-distribution within 2 rounds; the* H_0 *imprint is confined to round 0. Furthermore,* $\mathcal{W}(H_0) = 502$ *is* not *a member of* $\mathcal{C}_1^{\text{core}}$*, and under iteration converges to* C_2 *in 10 steps:* $502 \to \cdots \to 471 \in C_2$.

Remark 9.2 (Design Interpretation). *The* H_0 *values are derived from the fractional parts of the square roots of the first 8 primes ("nothing-up-my-sleeve" numbers). The* $\mathcal{W}(H_0) = 502$ *imprint is therefore not an intentional design choice but an arithmetic consequence of this construction. Its detectability through the CDP lens provides an additional lens for auditing SHA-256's initialization.*

10 Sequence Asymmetry

Observation 10.1 (Concatenation Order Variance Reduction). *For random equal-length strings* A, B *with* $A \neq B$*:*

$$\mathbb{E}[\mathcal{W}(\text{SHA256}(A\|B)) + \mathcal{W}(\text{SHA256}(B\|A))] \approx 960 \approx 2 \cdot \mathbb{E}[\mathcal{W}]$$

The mean is close to $2 \times 479.8 = 959.6$ *(the independent expectation). However, the standard deviation of the sum is:*

$$\sigma_{\mathcal{W}(A\|B)+\mathcal{W}(B\|A)} \approx 52 \quad \textit{vs.} \quad 2\sigma_{\mathcal{W}} = 73.6 \quad \textit{(if independent)}$$

The variance reduction factor is $52/73.6 \approx 0.71$*. Since* $A\|B$ *and* $B\|A$ *contain identical bytes in different positions, SHA-256's output* $\mathcal{W}$*-values are positively correlated. This correlation is measurable across all tested string lengths* ($L \in \{1, 2, 3, 4, 5, 6, 8\}$) *and constitutes a statistical footprint of SHA-256's non-commutative block processing.*

11 Application: CDP Rainbow Chains

11.1 Collision-Free Reduction Function

Classical Hellman rainbow tables suffer from chain merge probability proportional to $1 - e^{-t^2/2N}$ [2], limiting coverage to approximately 86% of the target space. Since the CDP fingerprint F is bijective over constrained input spaces, using F as the reduction function eliminates direct reduction collisions:

Theorem 11.1 (Merge-Free CDP Chains). *If* F *is injective over the target input space* X*, then CDP rainbow chains have zero* reduction-collision *probability. For a reduction function* R *based on the full CDP fingerprint, the per-step per-pair merge probability is bounded by the residual PCG string-generation collision rate* $\approx 1/|X|$.

Practical note (v2): *GPU implementations using a PCG-seeded string generator exhibit birthday-paradox merges at the rate expected for any surjective reduction (observed: 66.7% unique chains for* $n \times t/|X| = 1.0$*, vs. 36.8% for purely random reduction). The CDP fingerprint reduces the effective merge probability by* $\approx 1.81\times$ *compared to random, confirming structural benefit, but does not achieve the theoretical zero-merge bound. A true bijective reduction requires an explicit index-to-string mapping derived from* F.

11.2 Storage-Time Tradeoff

For n-character passwords over alphabet Σ, with chain length t and GPU rate R:

$$\text{Storage} = \frac{|\Sigma|^n}{t} \times 14 \text{ bytes}, \qquad \text{Query time} = \frac{t}{R}$$

Table 4: CDP chain table parameters (GPU rate $R = 2.875 \times 10^9$ H/s; observed rate: 4.6 GH/s on AMD RX 9070 XT).

Space	$\lvert\Sigma\rvert^n$	t	Storage	Build (single GPU)
8-char lowercase	2.1×10^{11}	10^6	3 MB	∼73 s
8-char alnum (62)	2.2×10^{14}	10^6	3 GB	∼3 h
8-char full (94)	6.1×10^{15}	10^6	85 GB	∼25 d
10-char lowercase	1.4×10^{14}	10^6	2 GB	∼14 h

12 Scope and Limitations

CDP does not break SHA-256. The Davies-Meyer construction ensures that $r(\iota(\text{state}^{(0)}), \mathcal{W}(\text{final hash})) = -0.033$ (empirically independent), confirming that intermediate basin states are uncorrelated with the final output. Preimage and collision resistance are unaffected.

CDP-based rainbow tables are effective against *unsalted* SHA-256 databases. Any application of a random per-user salt completely neutralizes CDP: the $\mathcal{W}$ value of SHA256(salt‖password) is uniformly distributed and independent of the structural properties described here. Modern password storage (bcrypt, scrypt, Argon2) is not affected.

13 Conclusion

We have presented CDP, a framework revealing structural properties of SHA-256's output distribution that are absent from the existing literature. The four proven theorems — complement nibble invariant, 20% M-rate convergence, universal collapse at padding boundaries, and ergodic Markov basin pressure — characterize SHA-256's behavior under the digit-sum projection in ways not predicted by its design specification.

Version 3 corrects the complement symmetry error of prior versions: hw_7 inputs show *elevated* C_1 affinity ($C_1\% = 21.9\%$) and hw_1 inputs show *suppressed* C_1 affinity ($C_1\% = 14.1\%$), opposite to the v1/v2 claim. The structural mechanism is identified: the Hamming weight of input bytes shifts the SHA-256 output $\mathcal{W}$-mean, and C_1-affinity tracks proximity to the $\mathcal{C}_1^{\text{core}}$ centre at 479.36. All other content is identical to v2.

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